Exercise C - I

1. Let $\frac{3\pi}{4} < \alpha < \pi$, then $\sqrt{2\cot\alpha + \frac{1}{\sin^2\alpha}}$ is equal to :

यदि $\frac{3\pi}{4} < \alpha < \pi$ तब $\sqrt{2\cot\alpha + \frac{1}{\sin^2\alpha}}$ किस के बराबर है :

(A) $\frac{\sin\alpha - \cos\alpha}{\sin\alpha}$ (B) $\frac{\sin\alpha + \cos\alpha}{\sin\alpha}$ (C) $\frac{-\sin\alpha + \cos\alpha}{\sin\alpha}$ (D) $\frac{-\sin\alpha - \cos\alpha}{\sin\alpha}$

Ans. D

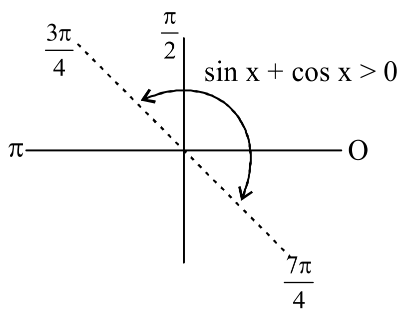
Sol. $\frac{3\pi}{4} < \alpha < \pi$

$$\sqrt{2\cot\alpha + \frac{1}{\sin^2\alpha}} = \frac{\sqrt{2\cot\alpha \cdot \sin^2\alpha + 1}}{|\sin\alpha|}$$

$$= \frac{\sqrt{1+\sin 2\alpha}}{\sin\alpha}$$

$$= \frac{|\sin\alpha + \cos\alpha|}{\sin\alpha}$$

$$= \frac{-\sin\alpha - \cos\alpha}{\sin\alpha} \qquad \left\{\because \frac{3\pi}{4} < \alpha < \pi\right\}$$



2. If $\cos^2 A + \cos^2 B + \cos^2 C = 1$, then triangle ABC is :

(A) right angled (B) equilateral (C) isosceles (D) None of these

यदि $\cos^2 A + \cos^2 B + \cos^2 C = 1$ तब त्रिभुज ABC एक :

(A) समकोण त्रिभुज होगा (B) समबाहु त्रिभुज होगा (C) समद्विबाहु त्रिभुज होगा (D) इनमें से कोई नहीं

Ans. A

Sol. $\cos^2A + \cos^2B + \cos^2C = 1$

$\Rightarrow 1 + \cos2A + 1 + \cos 2B + 1 + \cos2C = 2$

$\Rightarrow \cos2A + \cos2B + \cos2C = -1$

$\Rightarrow 2\cos(A + B)\cos(A - B) + \cos 2C = -1$

$\Rightarrow -2\cos(C)\cos(A - B) + 2\cos^2C = 0$

$\Rightarrow -2\cos C\,[\cos(A - B) - \cos C] = 0$

$\Rightarrow 2\cos C\,[\cos(A - B) + \cos(A + B)] = 0$

$\Rightarrow 4\cos C\cos A\cos B = 0$

$\Rightarrow A = \frac{\pi}{2}$ or $B = \frac{\pi}{2}$ or $C = \frac{\pi}{2}$

3. If $\frac{\tan 3A}{\tan A} = k$, then $\frac{\sin 3A}{\sin A}$ is equal to :

(A) $\frac{2k}{k-1}$ (B) $\frac{2k}{k+1}$ (C) $\frac{3k}{k+1}$ (D) None of these

यदि $\frac{\tan 3A}{\tan A} = k$, then $\frac{\sin 3A}{\sin A}$ किस के बराबर है :

(A) $\frac{2k}{k-1}$ (B) $\frac{2k}{k+1}$ (C) $\frac{3k}{k+1}$ (D) इनमें से कोई नहीं

Ans. A

Sol. $\frac{\tan 3A}{\tan A} = k$

Apply C & D rule

$\Rightarrow \frac{\tan 3A + \tan A}{\tan 3A - \tan A} = \frac{K+1}{K-1}$

$\Rightarrow \frac{\sin 3A \cos A + \cos 3A \sin A}{\sin 3A \cos A - \cos 3A \sin A} = \frac{K+1}{K-1}$

$\Rightarrow \frac{\sin 4A}{\sin 2A} = \frac{K+1}{K-1}$

$\Rightarrow 2\cos 2A = \frac{K+1}{K-1}$

$\Rightarrow \frac{\sin 3A}{\sin A} = \frac{3\sin A - 4\sin^3 A}{\sin A}$

$= 3 - 4\sin^2 A$

$= 3 - 4\left[\frac{1-\cos 2A}{2}\right]$

$= 3 - 2 + 2\cos 2A$

$= 1 + \frac{K+1}{K-1} = \frac{2K}{K-1}$

4. $3(\sin x - \cos x)^4 + 6(\sin x + \cos x)^2 + 4(\sin^6 x + \cos^6 x) =$

(A) 11 (B) 12 (C) 13 (D) 14

Ans. C

Sol. $3(\sin x - \cos x)^4 + 6(\sin x + \cos x)^2 + 4(\sin^6 x + \cos^6 x)$

$= 3[\sin^2 x + \cos^2 x - \sin 2x]^2 + 6[1 + \sin 2x] + 4[1 - 3\sin^2 x\cos^2 x]$

$\Rightarrow 3(\sin x - \cos x)^4 + 6(\sin x + \cos x)^2 + 4(\sin^6 x + \cos^6 x)$

$= 3[1 + 4\sin^2 x\cos^2 x - 4\sin x\cos x] + 6 + 12\sin x\cos x + 4 - 12\sin^2 x\cos^2 x$

$= 3 + 12\sin^2 x\cos^2 x - 12\sin x\cos x + 6 + 12\sin x\cos x + 4 - 12\sin^2 x\cos^2 x$

$= 13$

5. $\sec^2\theta = \dfrac{4xy}{(x+y)^2}$ is true if and only if :

$\sec^2\theta = \dfrac{4xy}{(x+y)^2}$ सत्य होगा यदि और केवल यदि :

(A) $x + y \neq 0$ (B) $x = y,\ x \neq 0$ (C) $x = y$ (D) $x \neq 0, y \neq 0$

Ans. B

Sol. $\sec^2\theta = \dfrac{4xy}{(x+y)^2}$

$\because \sec^2\theta \in [1,\infty)$

$\Rightarrow \dfrac{4xy}{(x+y)^2} \geq 1$

$\Rightarrow \dfrac{4xy - x^2 - y^2 - 2xy}{(x+y)^2} \geq 0$

$\Rightarrow \dfrac{-(x^2 + y^2 - 2xy)}{(x+y)^2} \geq 0$

$\Rightarrow \dfrac{(x-y)^2}{(x+y)^2} \leq 0$

$\Rightarrow x - y = 0\ \&\ x + y \neq 0$

$\Rightarrow x = y;\ x \neq 0$

6. If $\alpha + \beta = \dfrac{\pi}{2}$ and $\beta + \gamma = \alpha$, then $\tan\alpha$ equals :

यदि $\alpha + \beta = \frac{\pi}{2}$ एवं $\beta + \gamma = \alpha$ तब $\tan \alpha$ किस के बराबर होगा :

(A) $2(\tan \beta + \tan \gamma)$ (B) $\tan \beta + \tan \gamma$ (C) $\tan \beta + 2\tan \gamma$ (D) $2\tan \beta + \tan \gamma$

Ans. C

Sol. $\alpha + \beta = \frac{\pi}{2}$ & $\beta + \gamma = \alpha$

$\tan(\beta + \gamma) = \tan \alpha$

$\Rightarrow \frac{\tan \beta + \tan \gamma}{1 - \tan \beta \tan \gamma} = \tan \alpha$

$\Rightarrow \tan \beta + \tan \gamma = \tan \alpha - \tan \alpha \tan \beta \tan \gamma$

$\Rightarrow \tan \beta + \tan \gamma = \tan \alpha - \tan\left(\frac{\pi}{2} - \beta\right) \tan \beta \tan \gamma$

$\Rightarrow \tan \alpha = \tan \beta + 2\tan \gamma$

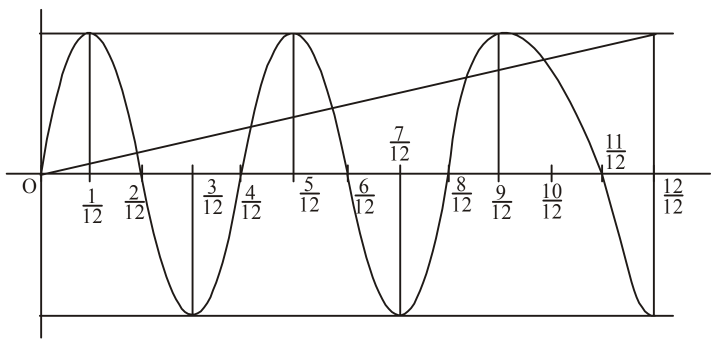
7. Number of real solutions to the equation $\sin(6\pi x) = x$, is :

समीकरण $\sin(6\pi x) = x$ के वास्तविक मानों की संख्या होगी :

(A) 13 (B) 11 (C) 9 (D) 7

Ans. B

Sol. Draw graphs of $y = x$ and $y = \sin 6\pi x$ and then interpret.



8. The value of x satisfying the equation, $x = \sqrt{2 + \sqrt{2 - \sqrt{2 + x}}}$ is :

समीकरण $x = \sqrt{2 + \sqrt{2 - \sqrt{2 + x}}}$ को सन्तुष्ट करने वाले x का मान होगा :

(A) $2\cos 10°$ (B) $2\cos 20°$ (C) $2\cos 40°$ (D) $2\cos 80°$

Ans. C

Sol. Note that $x \in [-2, 2]$ for x to be real
Let $x = 2\cos\theta$ where $\theta \in [0, \pi]$

$$x = \sqrt{2+\sqrt{2-\sqrt{2+2\cos\theta}}}$$

$$2\cos\theta = \sqrt{2+\sqrt{2-2\cos\frac{\theta}{2}}} = \sqrt{2+\sqrt{2\left(1-\cos\frac{\theta}{2}\right)}} = \sqrt{2+2\sin\frac{\theta}{4}} = \sqrt{2+2\cos\left(\frac{\pi}{2}-\frac{\theta}{4}\right)}$$

$$= \sqrt{2\left(1+\cos\left(\frac{\pi}{2}-\frac{\theta}{4}\right)\right)} = \sqrt{4\cos^2\left(\frac{\pi}{4}-\frac{\theta}{8}\right)}$$

$2\cos\theta = 2\cos \quad \Rightarrow \quad \theta = \frac{\pi}{4}-\frac{\theta}{8} \quad \Rightarrow \quad \frac{9\theta}{8} = \frac{\pi}{4} \quad \Rightarrow \quad \theta = \frac{2\pi}{9}$

Hence $x = 2\cos\frac{2\pi}{9} = 2\cos 40°$

9. The value of, (tan 18°)(sin 36°)(cos 54°)(tan 72°)(tan 108°)(cos 126°)(sin 144°)(tan162°)(cos180°) is k $\sin^2 18°$ then k has the value equal to :

यदि (tan 18°)(sin 36°)(cos 54°)(tan 72°)(tan 108°)(cos 126°)(sin 144°)(tan162°)(cos180°) = k$\sin^2$ 18° हो, तो k का मान होगा :

(A) $\frac{5}{16}$ (B) $\frac{5}{4}$ (C) $\frac{1}{4}$ (D) $\frac{1}{16}$

Ans. B

Sol. (tan 18°)(tan 72°)(tan 108°)(tan 162°) = 1
& sin36° cos54°cos(126°)sin(144°) = $\sin^4 36°$

$$\Rightarrow k\sin^2 18° = \left(\frac{1-\cos 72°}{2}\right)^2 = \frac{(1-\sin 18°)^2}{4}$$

$$= \frac{1}{4}\left(1-\frac{\sqrt{5}-1}{4}\right)^2 = \frac{1}{4}\left(\frac{5-\sqrt{5}}{4}\right)^2 = \frac{5}{4}\left(\frac{\sqrt{5}-1}{4}\right)^2 = \frac{5}{4}\sin^2 18° \Rightarrow k = \frac{5}{4}$$

10. If $x\sec\alpha + y\tan\alpha = x\sec\beta + y\tan\beta = a$, then $\sec\alpha \cdot \sec\beta =$

यदि $x\sec\alpha + y\tan\alpha = x\sec\beta + y\tan\beta = a$, तो $\sec\alpha \cdot \sec\beta =$

(A) $\frac{a^2+y^2}{x^2+y^2}$ (B) $\frac{a^2+y^2}{x^2-y^2}$ (C) $\frac{x^2+y^2}{a^2+y^2}$ (D) $\frac{x^2-y^2}{a^2-y^2}$

Ans. B

Sol. α and β satisfy the equation

$$x \sec\theta + y \tan\theta = a$$

$$(x\sec\theta - a)^2 = y^2\tan^2\theta = y^2(\sec^2\theta - 1)$$

$$\sec^2\theta\,(x^2 - y^2) - 2ax\sec\theta + a^2 + y^2 = 0$$

This is a quadratic in $\sec\theta$, whose roots are $\sec\alpha$ and $\sec\beta$

$$\Rightarrow \quad \sec\alpha \cdot \sec\beta = \frac{a^2 + y^2}{x^2 - y^2}$$

11. The graphs of $y = \sin x$, $y = \cos x$, $y = \tan x$ and $y = \text{cosec}\, x$ are drawn on the same axes from 0 to $\pi/2$. A vertical line is drawn through the point where the graphs of $y = \cos x$ and $y = \tan x$ cross, intersecting the other two graphs at points A and B. The length of the line segment AB is :

यदि $[0, \pi/2]$ के अन्तराल में $y = \sin x$, $y = \cos x$, $y = \tan x$ तथा $y = \text{cosec}\, x$ का आरेख खींचा जाता है। $y = \cos x$ एवं $y = \tan x$ के प्रतिच्छेद बिन्दु से y अक्ष के समान्तर एक रेखा खींची जाती है जो बाकी दो आरेख A एवं B पर काटती हो तो AB की लम्बाई क्या होगी :

(A) 1 (B) $\frac{\sqrt{5}-1}{2}$ (C) $\sqrt{2}$ (D) $\frac{\sqrt{5}+1}{2}$

Ans. A

Sol. $\tan x = \cos x$

$\Rightarrow \cos^2 x = \sin x$

$\Rightarrow 1 - \sin^2 x = \sin x$

$\Rightarrow \sin^2 x + \sin x - 1 = 0$

$$\Rightarrow \sin x = \frac{-1 \pm \sqrt{5}}{2}$$

$$\Rightarrow \sin x = \frac{\sqrt{5}-1}{2} \qquad \left\{\because 0 < x < \frac{\pi}{2}\right\}$$

$$\text{cosec}\, x = \frac{2}{\sqrt{5}-1}$$

$$\Rightarrow \left|\frac{2}{\sqrt{5}-1} - \frac{\sqrt{5}-1}{2}\right|$$

$$= \left|\frac{4-\left(5+1-2\sqrt{5}\right)}{2\left(\sqrt{5}-1\right)}\right|$$

$= 1$

12. The value of $4\cos\frac{\pi}{10} - 3\sec\frac{\pi}{10} - 2\tan\frac{\pi}{10}$ is equal to :

(A) 1 (B) $\sqrt{5}-1$ (C) $\sqrt{5}+1$ (D) zero

$4\cos\frac{\pi}{10} - 3\sec\frac{\pi}{10} - 2\tan\frac{\pi}{10}$ का मान होगा :

(A) 1 (B) $\sqrt{5}-1$ (C) $\sqrt{5}+1$ (D) शून्य

Ans. D

Sol. $4\cos18^0 - \frac{3}{\cos18^0} - 2\tan18^0$

$$\frac{4\cos^2 18^0 - 3 - 2\sin 18^0}{\cos18^0} = \frac{2(1+\cos36^0) - 2\sin18^0 - 3}{\cos18^0}$$

$$\Rightarrow \quad \frac{2(1+\cos36^0 - \sin18^0) - 3}{\cos18^0} = \frac{2\left(1+\frac{1}{2}\right) - 3}{\cos18^0} = 0$$

13. The value of $\cot x + \cot(60^o + x) + \cot(120^o + x)$ is equal to :

व्यंजक $\cot x + \cot(60^o + x) + \cot(120^o + x)$ का मान होगा :

(A) cot3x (B) tan3x (C) 3 tan3x (D) $\frac{3 - 9\tan^2 x}{3\tan x - \tan^3 x}$

Ans. D

Sol. $\cot x + \frac{\cos(60+x)}{\sin(60+x)} + \frac{\cos(x-60)}{\sin(x-60)}$

$$= \frac{\cos x}{\sin x} + \frac{\sin(2x)}{\sin(x+60)\sin(x-60)}$$

$$= \frac{\cos x}{\sin x} + \frac{8\sin x\cos x}{4\sin^2 x - 3} = \frac{4\sin^2 x\cos x - 3\cos x + 8\sin^2 x\cos x}{4\sin^3 x - 3\sin x}$$

$$= \frac{3[3\cos x - 4\cos^3 x]}{\sin 3x} = 3\cot 3x = \frac{3[1-3\tan^2 x]}{3\tan x - \tan^3 x}$$

14. The minimum value of the function $f(x) = \frac{\sin x}{\sqrt{1-\cos^2 x}} + \frac{\cos x}{\sqrt{1-\sin^2 x}} + \frac{\tan x}{\sqrt{\sec^2 x - 1}} + \frac{\cot x}{\sqrt{\operatorname{cosec}^2 x - 1}}$ as x varies over all numbers in the largest possible domain of $f(x)$ is :

x के सभी सम्भावित प्रान्त के लिये फलन $f(x) = \frac{\sin x}{\sqrt{1-\cos^2 x}} + \frac{\cos x}{\sqrt{1-\sin^2 x}} + \frac{\tan x}{\sqrt{\sec^2 x - 1}} + \frac{\cot x}{\sqrt{\operatorname{cosec}^2 x - 1}}$ का न्यूनतम मान होगा :

(A) 4 (B) – 2 (C) 0 (D) 2

Ans. B

Sol. Let $y = f(x) = \frac{\sin x}{|\sin x|} + \frac{\cos x}{|\cos x|} + \frac{\tan x}{|\tan x|} + \frac{\cot x}{|\cot x|}$

$$\left.\begin{array}{l} \text{if } x \in 1^{st} \text{ quadrant then } y = 1+1+1+1 = 4 \\ \text{if } x \in 2^{nd} \text{ quadrant then } y = 1-1-1-1 = -2 \\ \text{if } x \in 3^{rd} \text{ quadrant then } y = -1-1+1+1 = 0 \\ \text{if } x \in 4^{th} \text{ quadrant then } y = -1+1-1-1 = -2 \end{array}\right\} \Rightarrow y_{min} = -2$$

15. If $a > b > 0$, then the minimum value of $a \sec\theta - b \tan\theta \ \forall \theta \in \left(0, \frac{\pi}{2}\right)$:

यदि $a > b > 0$ तो $\forall \theta \in \left(0, \frac{\pi}{2}\right)$ के लिये $a \sec\theta - b \tan\theta$ का न्यूनतम मान होगा :

(A) $\sqrt{a^2 - b^2}$ (B) $a - b$ (C) $\sqrt{a - b}$ (D) $(a^2 - b^2)^{3/2}$

Ans. A

Sol. $y + b \tan \theta = a \sec \theta$

$y^2 + 2by \tan \theta + b^2 \tan^2\theta = a^2 + a^2 \tan^2\theta$

$(b^2 - a^2)\tan^2\theta + 2by \tan \theta + (y^2 - a^2) = 0$

$D \geq 0$

$4b^2y^2 - 4(b^2 - a^2)(y^2 - a^2) \geq 0$

$b^2y^2 - [b^2y^2 - b^2a^2 - a^2y^2 + a4] \geq 0$

$b^2 + y^2 - a^2 \geq 0$

$y^2 \geq a^2 - b^2$

ONE OR MORE THAN ONE CORRECT TYPE QUESTIONS

16. If $\sin(x-y) = \cos(x+y) = \frac{1}{2}$, then the values of 'x' and 'y' lying between 0° and 180° are given by :

यदि $\sin(x-y) = \cos(x+y) = \frac{1}{2}$ तब 'x' एवं 'y' के वे मान जो 0° एवं 180° के बीच में हो होंगे :

(A) $x = 45°, y = 15°$ (B) $x = 45°, y = 135°$ (C) $x = 165°, y = 15°$ (D) $x = 165°, y = 135°$

Ans. AD

Sol. $\sin(x-y) = \cos(x+y) = \frac{1}{2}$

$\sin x \cos y - \cos x \sin y = \cos x \cos y - \sin x \sin y$

$\Rightarrow (\sin x - \cos x)\cos y = \sin y(\cos x - \sin x)$

$\Rightarrow (\cos y + \sin y)(\sin x - \cos x) = 0$

$\Rightarrow \sin x = \cos x$ or $\cos y = -\sin y$

$\Rightarrow x = \frac{\pi}{4}$ or $y = \frac{3\pi}{4}$

If $x = \frac{\pi}{4}$ then $\sin\left(\frac{\pi}{4} - y\right) = \frac{1}{2} \Rightarrow y = 15^{o}$

If $y = \frac{3\pi}{4}$ then $\sin\left(x - \frac{3\pi}{4}\right) = \frac{1}{2} \Rightarrow x = 165^{o}$

17. For a positive integer 'n', let : $f_n(\theta) = \left(\tan\frac{\theta}{2}\right)(1+\sec\theta)(1+\sec 2\theta)(1+\sec 4\theta) \ldots\ldots (1+\sec 2^n\theta)$. Then :

किसी धनात्मक पूर्णांक 'n' के लिए माना $f_n(\theta) = \left(\tan\frac{\theta}{2}\right)(1+\sec\theta)(1+\sec 2\theta)(1+\sec 4\theta) \ldots\ldots (1+\sec 2^n\theta)$ तब:

(A) $f_2\left(\frac{\pi}{16}\right) = 1$ (B) $f_3\left(\frac{\pi}{32}\right) = 1$ (C) $f_4\left(\frac{\pi}{64}\right) = 1$ (D) $f_5\left(\frac{\pi}{128}\right) = 1$

Ans. ABCD

Sol. $f_n(\theta) = \left(\tan\frac{\theta}{2}\right)(1+\sec\theta)(1+\sec 2\theta)\ldots\ldots\ldots\ldots(1+\sec 2^n\theta)$

$\Rightarrow f_n(\theta) = \left(\tan\frac{\theta}{2}\right)\left(\frac{1+\cos\theta}{\cos\theta}\right)\cdot\left(\frac{1+\cos 2\theta}{\cos 2\theta}\right)\ldots\ldots\ldots\ldots\left(\frac{1+\cos 2^n\theta}{\cos 2^n\theta}\right)$

$$\Rightarrow f_n(\theta)=\frac{\sin\frac{\theta}{2}}{\cos\frac{\theta}{2}}\cdot\frac{2\cos^2\frac{\theta}{2}}{\cos\theta}\cdot\frac{2\cos^2\theta}{\cos 2\theta}\cdot\frac{2\cos^2 2\theta}{\cos 4\theta}\cdot\frac{..........}{\cos 2^{n-1}\theta}\cdot\frac{2\cos^2 2^{n-1}\theta}{\cos 2^n\theta}$$

$= \tan 2^n\theta$

$$f_2\left(\frac{\pi}{16}\right)=\tan\left(4\cdot\frac{\pi}{16}\right)=1$$

$$f_3\left(\frac{\pi}{32}\right)=\tan\left(8\cdot\frac{\pi}{32}\right)=1$$

$$f_4\left(\frac{\pi}{64}\right)=\tan\left(16\cdot\frac{\pi}{64}\right)=1$$

$$f_5\left(\frac{\pi}{128}\right)=\tan\left(\frac{32\pi}{128}\right)=1$$

18. If S is the solution set of all values of x for which $2\cos t=\frac{3x+1}{x-1}$, then :

(A) number of integral values in S is 5
(B) number of integral values in S is 4.
(C) sum of all integral values in S is –6
(D) sum of all integral values in S is –5

यदि $2\cos t=\frac{3x+1}{x-1}$ को संतुष्ट करने वाले x के हलों का समुच्चय S है, तो :

(A) S में पूर्णांक मानों की संख्या 5 है
(B) S में पूर्णांक मानों की संख्या 4 है
(C) S में पूर्णांक मानों का योग –6 है
(D) S में पूर्णांक मानों का योग –5 है

Ans. BC

Sol. $-2 \le 2\cos t \le 2$

$$-2 \le \frac{3x+1}{x-1} \le 2$$

$$\Rightarrow \frac{3x+1}{x-1}+2\ge 0 \;\;\&\;\; \Rightarrow \frac{3x+1}{x-1}-2\le 0$$

$$\Rightarrow \frac{5x+1}{x-1}\ge 0 \;\;\&\;\; \frac{x+3}{x-1}\le 0$$

$$\Rightarrow x\in\left[-3,-\frac{1}{5}\right]\Rightarrow x=-3,-2,-1,0$$

19. Let $f_n(\theta) = 2(\sin^2\theta + \sin 2\theta \sin 4\theta + \sin 3\theta \sin 9\theta + ... + \sin n\theta \sin n^2\theta)$ then which of the following hold(s) good :

माना कि $f_n(\theta) = 2(\sin^2\theta + \sin 2\theta \sin 4\theta + \sin 3\theta \sin 9\theta + ... + \sin n\theta \sin n^2\theta)$, तो निम्न में से कौनसे कथन सत्य

है :

(A) $f_3\left(\frac{\pi}{6}\right)=0$ (B) $f_4\left(\frac{\pi}{3}\right)=2$ (C) $f_8\left(\frac{\pi}{2}\right)=0$ (D) $f_{11}\left(\frac{\pi}{4}\right)=2$

Ans. ACD

Sol. $f_n(\theta)=\sum_{r=1}^{n}2\sin r\theta\,\sin r^2\theta$

$=\sum_{r=1}^{n}\left[\cos(r^2-r)\theta-\cos(r^2+r)\theta\right]$

$=\cos 0^{o}-\cos(n^2+n)\theta=1-\cos(n^2+n)\theta$

$\therefore\ f_3\left(\frac{\pi}{6}\right)=1-\cos(9+3)\frac{\pi}{6}=1-\cos 2\pi=0$

$f_8\left(\frac{\pi}{2}\right)=1-\cos(64+8)\frac{\pi}{2}=1-\cos 36\pi=0$

$f_{11}\left(\frac{\pi}{4}\right)=1-\cos(121+11)\frac{\pi}{4}=1-\cos 33\pi=2$

$f_4\left(\frac{\pi}{3}\right)=1-\cos\left(\frac{19\pi}{3}\right)=1-\cos\frac{\pi}{3}=\frac{1}{2}$